Adenosine

Mechanism of action

- Adenosine is an antidysrhythmic used for the treatment of paroxysmal supraventricular tachycardia (SVT).
- Adenosine is a nucleoside that depresses conduction through the AV node. This interrupts re-entry circuits within the heart and may restore sinus rhythm in patients with SVT.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Patients aged greater than or equal to 12 years with SVT and a ventricular rate greater than or equal to 150/minute, and
  - The patient is not severely compromised, and
  - The rhythm fails to revert following two Valsalva manoeuvres.

Contraindications

- ✕ Known severe allergy.
- ✕ Known sick sinus syndrome without an internal pacemaker in place. Adenosine may cause severe bradycardia if the patient has sick sinus syndrome.
- ✕ Previous 2nd or 3rd degree heart block without an internal pacemaker in place. Adenosine may cause heart block if the patient has had previous heart block.
- ✕ Previous heart transplantation without an internal pacemaker in place. Following a heart transplant the heart is denervated and adenosine may cause severe bradycardia.

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Cautions

- Asthma. Adenosine may precipitate bronchospasm and should be withheld if the patient has had recurrent life-threatening attacks of bronchospasm, or is currently suffering an exacerbation of asthma.
- COPD. Adenosine may precipitate bronchospasm and should be withheld if the patient has had recurrent life-threatening attacks of bronchospasm, or is currently suffering an exacerbation of COPD.
- Wolff-Parkinson-White (WPW) syndrome if the rhythm is possibly fast atrial fibrillation. Adenosine is not contraindicated in a patient with known WPW syndrome provided the rhythm is clearly SVT. If the rhythm is possibly fast atrial fibrillation, adenosine should be withheld because of the risk of precipitating VF.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. However, the balance of risk is in favour of administration if indicated.

Dosage

- 6 mg.
- A second dose of 12 mg may be administered if the rhythm fails to revert.

Administration

- The preferred site of injection is the antecubital fossa.
- Administer the dose as a rapid bolus followed immediately by a rapid flush of a minimum of 20 ml of 0.9% sodium chloride.

Common adverse effects

- Bradycardia and/or sinus pause which may be up to 30 seconds.
- Shortness of breath and/or an urge to breathe deeply.
- Feeling light-headed.
- Nausea and flushing.
- Feeling of chest pressure and/or severe apprehension.

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Usual onset of effect

- 5-10 seconds.

Usual duration of effect

- 10-20 seconds.

Pharmacokinetics

- Adenosine is rapidly taken up and metabolised within seconds by red blood cells and vascular endothelial cells.

Usual preparation

- Ampoule containing 6 mg in 2 ml.

Common interactions

- None.

Additional information

- Adenosine usually causes a brief period of very low cardiac output and this often causes the patient to feel severe apprehension and/or an impending sense of doom. Warn the patient they may feel awful but reassure them this will pass very quickly.
- If the rhythm is possibly fast atrial fibrillation and an accessory pathway is present, adenosine may precipitate VF. If uncertain, treat the rhythm as atrial fibrillation and not SVT.
- Seek clinical advice if the patient is aged less than 12 years.

More Information

Version history

11 September 2023 1.0.4

Update: Indications updated. Addition to administration and additional information headings.

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